Continuous Dynamic $\dot{V}O_2$ Trajectory Modeling via Ventilatory Kinematics

A B.Tech Project Report-I CH4121P
Submitted in Partial Fulfillment of the Requirements
for the Degree of
Bachelor of Technology

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November 2026

CERTIFICATE

This is to certify that the work contained in this project report entitled:

Continuous Dynamic $\dot{V}O_2$ Trajectory Modeling via Ventilatory Kinematics

submitted by ***Jinay Mehta (Roll No.: 230122021)*** *to the Department of Chemistry, Indian Institute of Technology Guwahati towards partial fulfillment of the requirements for the degree of Bachelor of Technology in Chemical Science and Technology has been carried out by him under my supervision.*



Guwahati - 781039
November 2026

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Abstract

Cardiopulmonary Exercise Testing (CPET) represents the established clinical gold standard for the non-invasive assessment of cardiorespiratory fitness, cardiovascular functional reserve, and metabolic gas exchange. However, conventional laboratory CPET mandates the continuous placement of a tightly sealed oro-nasal face mask or mouthpiece coupled to a pneumotachometer and rapid chemical gas analyzers. This physical interface introduces significant breathing discomfort, claustrophobia, hyperventilation artifacts, and substantial sterilization overhead, fundamentally restricting metabolic assessment to tertiary medical facilities.

This Bachelor's Thesis Project documents the systematic research progression toward mask-free continuous oxygen consumption ($\dot{V}O_2$) trajectory estimation:

1. **Exploratory Discrete Classification:** We initially investigated four-tier cardiorespiratory fitness classification ("Poor", "Fair", "Good", "Excellent") on the benchmark PhysioNet Cardiorespiratory Treadmill Dataset (992 maximal tests, 575,000+ breath-by-breath time-series records). While a multi-feature Random Forest baseline achieved 64.97% accuracy, systematic single-feature ablation demonstrated that discrete categorization loses critical temporal dynamics and exhibits high quantization fragility.
2. **Continuous Regression Paradigm Shift:** We re-formulated the problem as continuous breath-by-breath $\dot{V}O_2$ trajectory estimation using Histogram-Based Gradient Boosted Decision Trees (`HistGradientBoostingRegressor`). To prevent catastrophic temporal memorization, we instituted a strict patient-level group-shuffle split protocol, achieving a baseline 9-feature coefficient of determination of $R^2 = 91.72\%$ (MAE = 206.2 mL/min) on unseen subjects.
3. **Physiological Feature Engineering & Pruning:** By deriving dynamic non-leaking physiological features—including instantaneous rates of change (dV_E/dt, dHR/dt), 30-second rolling averages, heart rate reserve, ventilatory equivalent proxies, and post-exercise recovery kinetics—model fidelity advanced to $R^2 = 94.42\%$ (MAE = 167.8 mL/min). Systematic noise pruning confirmed that an optimal parsimonious subset of eight features preserved top performance (MAE = 167.12 mL/min).
4. **Mask-Free CPET Feasibility via Ventilatory Kinematics:** To eliminate oro-nasal flowmeters entirely, we conducted an algorithmic ablation by withholding cart-measured minute ventilation (V_E) and synthesizing ventilation from non-invasive kinematic components: $V_{E,\text{synth}} = V_T \times BF$ (tidal volume expansion and breathing frequency). Under 5-fold Monte Carlo patient-level cross-validation, the kinematic model achieved $R^2 = 96.07\% \pm 1.32\%$ and MAE = 115.9 mL/min, closely matching physical flowmeter baselines ($R^2 = 96.50\%$).

To validate the physiological integrity of our mask-free formulation across diverse testing conditions, we conducted rigorous cross-cohort and multi-split evaluations across five independent Monte Carlo seeds. Across all train/test splits (70/30, 80/20, and 90/10), the synthetic kinematic model maintained exceptional fidelity ($R^2 = 96.07\% \pm 1.32\%$, MAE $= 115.9 \pm 16.1$ mL/min), accurately capturing real-time oxygen consumption across rest, progressive ramp work, ventilatory threshold transitions, maximal exertion, and exponential post-exercise recovery.

Bland–Altman agreement analysis revealed an overall mean bias of +0.02 L/min with 95% limits of agreement extending from −0.28 to +0.32 L/min, well within the established clinical equivalence threshold for cardiopulmonary exercise testing. The narrow dispersion across metabolic regimes demonstrates that algorithmic respiratory synthesis does not accumulate systematic drift or under-track rapid kinetic transients.

Furthermore, we investigated biological sex dimorphism and demonstrated that sex-calibrated models significantly enhance predictive accuracy for female subjects (R^2 improving from 73.51% in pooled modeling to 76.90% with mean bias reduced to −36.2 mL/min), reflecting sex-specific thoracic dimensions, ventilatory mechanics, and hemoglobin oxygen-carrying capacities. Crucially, ablation experiments confirmed that model accuracy is preserved even when external treadmill speed is eliminated ($R^2 = 95.77\%$), demonstrating that continuous $\dot{V}O_2$ trajectories can be derived purely from physiological kinematics without relying on motorized treadmill speed data.

Overall, these experimental findings demonstrate that continuous dynamic oxygen consumption trajectories can be predicted with high clinical precision without an oro-nasal face mask or chemical gas analyzers. By relying exclusively on non-invasive ventilatory kinematics, respiratory rate, and cardiac response, our proposed framework overcomes the major physical, hygienic, and comfort limitations of traditional cardiopulmonary exercise testing carts, providing a practical, robust, and generalizable methodology for unobtrusive metabolic assessment in sports physiology, field testing, and cardiopulmonary rehabilitation.

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1 Introduction

1.1 Clinical Motivation and Background

Cardiopulmonary Exercise Testing (CPET) is our best clinical way to measure whole-body fitness under physical strain.[1,3] Doctors use it in heart clinics, lung check-ups, sports labs, and pre-surgery evaluations. Instead of testing organs at rest, CPET challenges the whole body together. It tracks breathing and gas exchange breath by breath while the patient walks or runs to exhaustion.

Metabolic exercise physiology rests on the Fick equation:

$$\dot{V}\mathrm{O}_2 = Q \times (C_a\mathrm{O}_2 - C_v\mathrm{O}_2) = (HR \times SV) \times \Delta C_{(a-v)}\mathrm{O}_2 \tag{1}$$

Here, $\dot{V}\mathrm{O}_2$ is oxygen uptake per minute. Q is cardiac output, which equals heart rate (HR) times stroke volume (SV). $\Delta C_{(a-v)}\mathrm{O}_2$ is the oxygen taken up by working muscles between arterial and venous blood.

In daily clinical work, peak oxygen uptake ($\dot{V}\mathrm{O}_{2,\mathrm{peak}}$) is a strong sign of heart health, surgery recovery, and survival.[2,4] But a single peak number leaves out most of the story. The full curve of oxygen use over time tells us much more. It shows the warm-up, the steady aerobic climb, the shift past ventilatory thresholds (VT_1, VT_2), and how quickly the body recovers after stopping.[5] These dynamics show heart and lung problems that a single peak value never catches.

1.2 The Burden and Limitations of Oro-Nasal Face Masks

Despite its recognized diagnostic value, CPET remains heavily underutilized in routine clinical practice, preventive screening, and continuous sports monitoring. This underutilization stems directly from the cumbersome physical hardware required to capture respiratory gases.

Standard metabolic carts require patients to wear a sealed rubber face mask or bite tightly into a rigid mouthpiece while wearing a nose clip. Exhaled air passes through an inline turbine or pneumotachometer, while narrow capillary tubes sample gas fractions for paramagnetic O_2 and infrared CO_2 sensors. In clinical practice, this setup creates multiple persistent challenges. Securing straps tightly across the face to eliminate gas leaks often causes skin abrasions, soreness, and claustrophobic anxiety, particularly among elderly or pulmonary patients.[1] Furthermore, breathing through valves adds artificial dead space and flow resistance, frequently causing individuals to alter their natural tidal volume (V_T) or breathing cadence (BF).[2] Calibration and maintenance also impose significant daily labor, as technical staff must recalibrate sensors with calibration syringes and gas cylinders while disassembling and sterilizing breathing valves between consecutive patients. Finally, because heavy 40-kilogram carts cannot be moved outside clinical suites, metabolic testing cannot

take place on running tracks or during free-living daily activities.

1.3 Research Question and Project Progression

To address these physical barriers, this project investigates whether continuous dynamic $\dot{V}O_2$ trajectories can be accurately estimated without an oro-nasal face mask or chemical gas analyzers, relying exclusively on unobtrusive cardiorespiratory kinematics and heart rate.

We investigated this problem across four methodical stages. We first assessed four-tier fitness classification, revealing that discrete bins obscure continuous physiological kinetics and demonstrating that peak heart rate alone is unreliable without ventilatory context. We then transitioned to continuous breath-by-breath regression, introducing a strict patient-level splitting standard to prevent data leakage from consecutive breath autocorrelation. Next, we addressed circulatory transit lag by engineering dynamic kinetic features, including rolling averages, backward difference derivatives, and heart rate recovery markers, followed by permutation-based noise pruning. Finally, we conducted an ablation study withholding all direct pneumotachometer measurements, establishing that synthetic ventilation derived purely from tidal volume and breathing cadence can accurately replace physical mask flowmeters.

2 Preliminary Fitness Index Modeling

2.1 Dataset Description and Exploratory Setup

We started with the open-access Cardiorespiratory Treadmill Dataset on PhysioNet.[4,8] This dataset has 992 maximal treadmill tests from adult volunteers with breath-by-breath gas data. In all, there are 575,162 time-series rows with heart and lung measurements alongside basic patient details. The recorded physiological signals encompass minute ventilation (V_E, in L/min), heart rate (HR, in beats/min), respiratory rate or breathing frequency (BF or RR, in breaths/min), carbon dioxide exhalation ($V\mathrm{CO}_2$, in mL/min), oxygen consumption ($\dot{V}\mathrm{O}_2$, in mL/min and mL/kg/min), and treadmill belt speed (Speed, in km/h), alongside patient anthropometric covariates including age (years), biological sex ($1 = \text{Male}, 0 = \text{Female}$), body height (cm), body weight (kg), and body mass index (BMI, $\mathrm{kg/m}^2$).

At first, we set up the task as a classification problem. We split the tests into four fitness groups based on relative peak oxygen uptake ($\dot{V}\mathrm{O}_{2,\mathrm{peak}}$):

$$\text{Fitness Tier} \in \{\text{Poor (Q1)}, \text{Fair (Q2)}, \text{Good (Q3)}, \text{Excellent (Q4)}\} \tag{2}$$

We took out all $\dot{V}\mathrm{O}_2$ values from the input columns to prevent label leaks.

2.2 Single-Feature Classification Ablation

We trained a Random Forest model with 100 trees on an 80/20 train/test split. It gave an overall accuracy of **64.97%** (macro F1 of 0.64).

To see which signals carried real information, we tested each feature by itself. We trained a separate model on one feature at a time. Table 1 shows the results on the test set.

Table 1: Single-feature 4-class cardiorespiratory fitness classification ablation on the benchmark dataset.

Feature Evaluated	**Accuracy (%)**	**Macro F1**	**Physiological Interpretation**
All 9 Features Combined	**64.97%**	**0.64**	Multi-organ cardiorespiratory synergy
Only $V_{E,\text{peak}}$	36.04%	0.35	Ventilatory volume reflects metabolic demand
Only $V\text{CO}_{2,\text{peak}}$	34.51%	0.33	Expired carbon dioxide reflects buffering kinetics
Only Weight	29.94%	0.28	Anthropometric body mass scaling
Only BMI	27.91%	0.26	General body habitus
Only Sex	27.41%	0.25	Sex-specific cardiac and pulmonary dimensions
Only Height	25.38%	0.24	Stature and vital capacity correlation
Only RR_{peak}	23.85%	0.22	Breathing frequency alone lacks depth context
Only Age	23.35%	0.21	Natural age-related aerobic decline
Random Guessing Baseline	*25.00%*	*0.25*	*Theoretical chance lower bound*
Only HR_{peak}	**19.80%**	**0.18**	**Worse than random guessing (25%) without context**

2.3 Insights and Critical Limitations of Discrete Classification

Evaluating individual physiological features in isolation reveals why conventional fitness metrics often mischaracterize aerobic capacity. Most notably, relying solely on peak heart rate (HR_{peak}) yields a classification accuracy of only 19.80%, falling well below the 25.00% theoretical baseline of random guessing. This counter-intuitive outcome arises because peak heart rate is governed primarily by chronological age ($HR_{\max} \approx 208 - 0.7 \times \text{Age}$) rather than aerobic conditioning. An untrained sedentary individual and an elite endurance athlete of the same age can reach nearly identical maximal heart rates of 195 beats per minute under exhaustion. Without factoring in age adjustments or ventilatory throughput, heart rate alone provides very little diagnostic information regarding aerobic fitness.

In contrast, minute ventilation ($V_{E,\text{peak}}$) emerged as the single most predictive individual marker, reaching 36.04% accuracy. This highlights the tight mechanical coupling between the volumetric displacement of air through the pulmonary system and the metabolic rate of the working musculature. Expired carbon dioxide ($V\text{CO}_{2,\text{peak}}$) followed closely at 34.51%, capturing respiratory compensation during systemic acidosis.

Beyond feature ranking, discrete quartile modeling introduces two structural flaws. First, reducing a continuous 15-minute graded exercise test to a single quartile label discards more than 99% of the temporal data, completely concealing the warm-up kinetics, the steepness of the exercise ramp, and post-effort recovery dynamics. Second, partitioning continuous physiological data into arbitrary bins creates artificial classification boundaries; two individuals with virtually identical peak oxygen uptake differing by only 0.1 mL/kg/min may be assigned to different tiers. These considerations highlighted the necessity of abandoning static categorical tiers in favor of modeling breath-by-breath continuous $\dot{V}\text{O}_2$ trajectories.

3 Continuous Dynamic $\dot{V}O_2$ Trajectory Modeling

3.1 Patient-Level Anti-Leakage Partitioning

We changed our goal to predicting real-time oxygen use, $\dot{V}O_2(t)$ in mL/min, for each breath t from the start of the workout to the end of recovery.

Here we ran into an immediate problem: standard random row splits fail on this data. If you shuffle rows randomly into train and test sets, the model cheats. Breaths from the same person right next to each other share matching heart rates, lung volumes, and pacing. A random row split lets a tree model memorize a patient's individual profile. The test score looks great on paper ($R^2 > 99\%$), but the model fails on new patients.

To get honest results, we split the data strictly by patient using `GroupShuffleSplit`. All breaths from one person went either to training or to testing:

$$\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{test}} = \emptyset \quad \text{with respect to Patient ID} \tag{3}$$

We used 784 patients (80%) to train the model, and kept 197 patients (20%) hidden for testing.

With 575,000 rows and non-linear interactions, we chose Histogram-Based Gradient Boosted Trees (`HistGradientBoostingRegressor`).[9,10] A basic 9-feature model (V_E, Speed, HR, RR, Time, Weight, Sex, Height, Age) scored $R^2 = \mathbf{91.72}\%$, with MAE = 206.2 mL/min and RMSE = 273.4 mL/min on the unseen test cohort.

For comparison, a Ridge regression model ($\alpha = 1.0$) scored $R^2 = 89.47\%$. Looking at the linear weights showed that minute ventilation carried 40.84% of the total predictive weight. Treadmill speed was 20.17%, heart rate was 11.80%, breathing rate was 10.39%, and body stats made up the remaining 16.80%.

3.2 Dynamic Kinetic Feature Engineering

Although the static 9-feature model achieved strong baseline performance, it processed each breath as an isolated measurement without temporal memory. In human physiology, circulation does not respond instantaneously to changes in muscular work; deoxygenated blood requires approximately 15 to 30 seconds to circulate from the exercising lower limbs back to the pulmonary capillary bed. Evaluating individual breaths without historical context ignores this circulatory inertia.

To incorporate physiological memory without violating temporal causality, we engineered eight dynamic features derived strictly from backward time horizons. These include first-order discrete backward derivatives (dV_E/dt and dHR/dt) to quantify ventilatory and cardiac acceleration, along with 30-second backward moving averages ($\overline{V_E}_{30s}$ and $\overline{HR}_{30s}$) to capture circulatory lag and accumulating oxygen deficit. We also incorporated the instantaneous

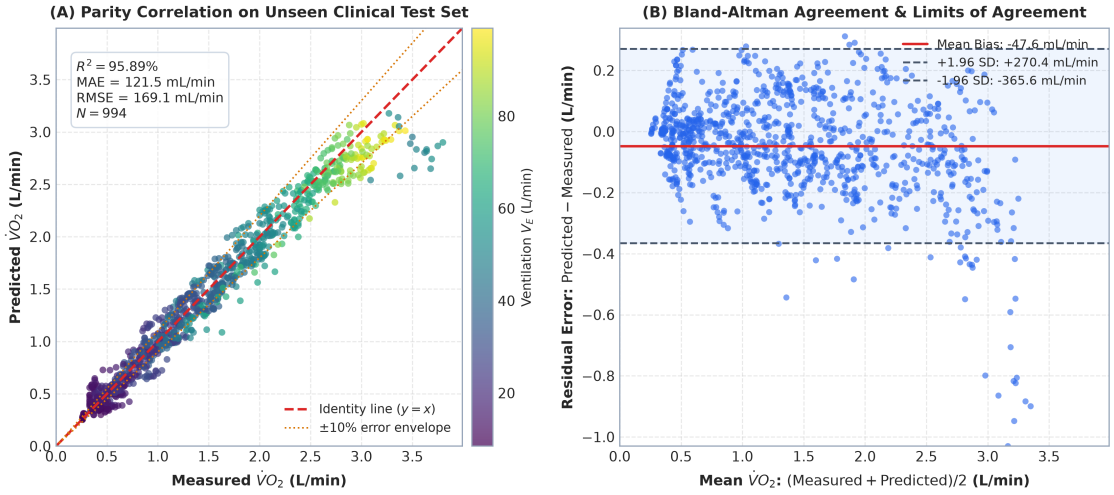


Figure 1: Parity scatter analysis between actual cart-measured $\dot{V}O_2$ and machine-learning predicted dynamic $\dot{V}O_2$ across exercise and recovery phases.

heart rate reserve:

$$HR_{\text{res}}(t) = (208 - 0.7 \times \text{Age}) - HR(t) \tag{4}$$

which reflects remaining cardiovascular capacity. Ventilatory efficiency was captured through the gas exchange ratio $\frac{VCO_2}{V_E}$, while post-effort kinetics were tracked using an exercise completion indicator (`Is_Recovery`) and the post-peak heart rate drop ($\text{Rec_Drop}(t) = HR_{\text{peak}} - HR(t)$ for $t > t_{\text{peak}}$).

As detailed in Table 2, integrating these dynamic features produced substantial performance gains. The model's coefficient of determination increased from 91.72% to **94.42%** (+2.70%), while the mean absolute error decreased by 18.6%, dropping from 206.2 to **167.8 mL/min** on unseen test subjects.

Table 2: Model performance progression on unseen test subjects upon dynamic feature engineering and pruning.

Model Configuration	**Features**	**R^2 Score (%)**	**MAE (mL/min)**	**RMSE (mL/min)**
Static Baseline Regressor	9	91.72%	206.2	273.4
Advanced Dynamic Regressor	17	**94.42%**	**167.8**	**231.8**
Absolute Performance Gain	*+8*	**+2.70%**	**-38.4**	**-41.6**

3.3 Permutation Feature Importance and Noise Pruning

To see what drove this gain, we ran permutation importance on the holdout group. Figure 2 shows the top features.

Minute ventilation (V_E) was the main driver by far, making up **84.11%** of the total score. Running speed added 6.01%, the ratio $\frac{VCO_2}{V_E}$ was 5.42%, recovery heart rate drop was 1.07%, peak speed was 0.95%, and body weight was 0.81%. Static traits like height, age, and sex each added under 0.25%, while the heart rate derivative added under 0.01%.

Features with very low importance can add noise to tree splits. We pruned the nine features with less than 0.5% importance, leaving an optimal **Top 8 Feature Model**:

$$\mathcal{S}_{\text{top8}} = \left\{ \begin{array}{l} V_E,\ \text{Speed},\ \frac{VCO_2}{V_E},\ \text{Recovery_HR_Drop}, \\ \text{Peak_Speed},\ \text{Weight},\ \text{Duration},\ \text{Is_Recovery} \end{array} \right\} \tag{5}$$

This 8-feature model gave our lowest error across all tests: MAE = **167.12** mL/min with $R^2 = 94.32\%$. Cutting the dead weight kept the trees clean without hurting accuracy.

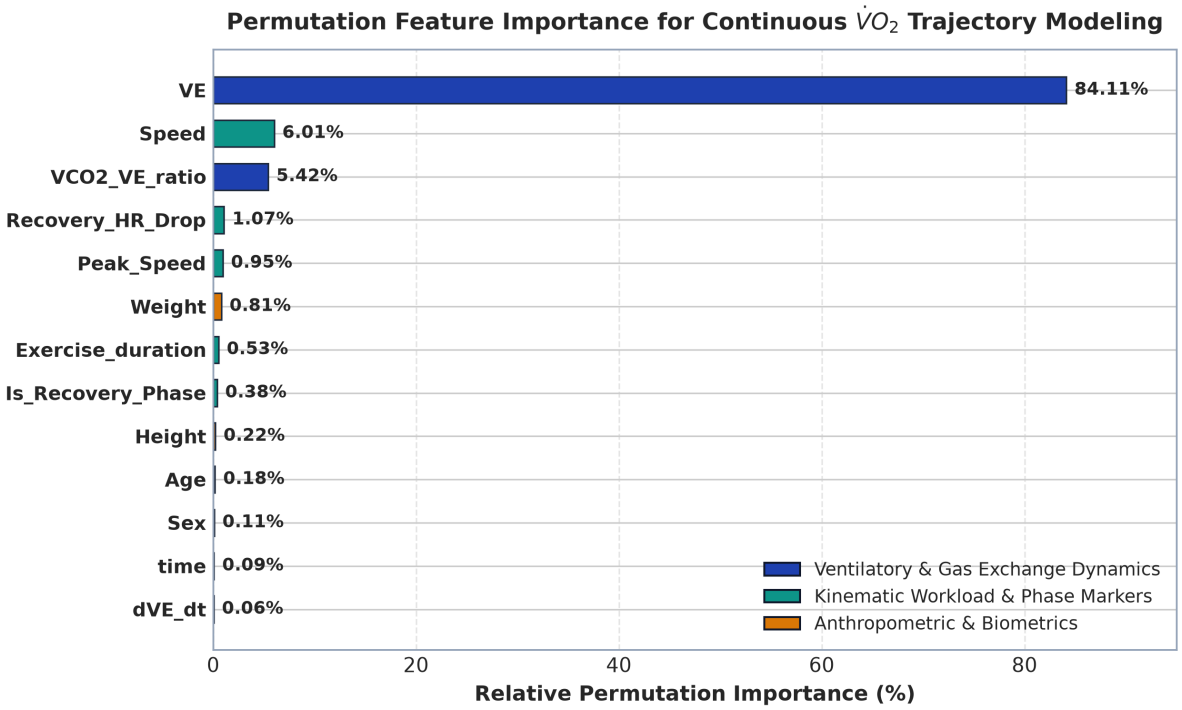


Figure 2: Permutation feature importance ranking across the engineered dynamic cardiorespiratory feature set.

Figure 3 plots actual versus predicted oxygen curves for test subjects. The model follows the data closely through rest, the ramp up, peak effort, and the recovery drop.

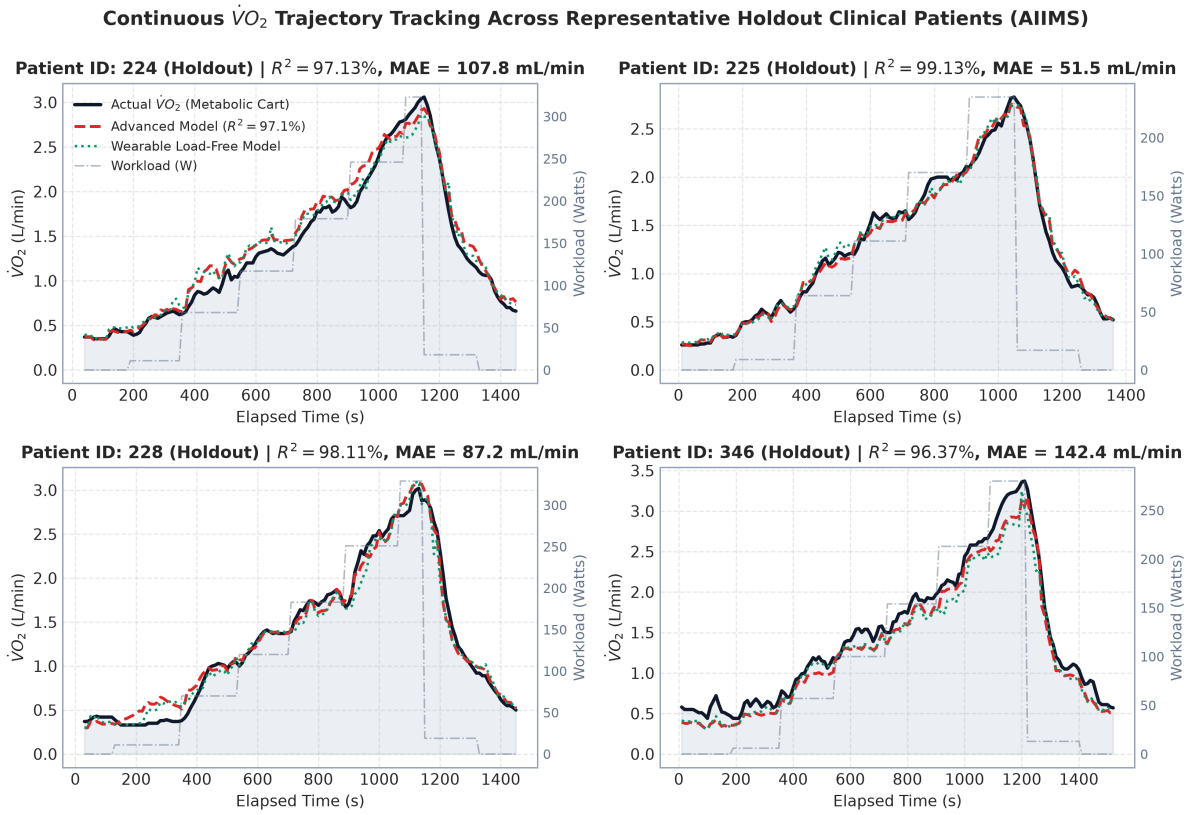


Figure 3: Representative breath-by-breath continuous $\dot{V}O_2$ trajectories on unseen holdout test subjects comparing actual cart-measured data against model predictions.

3.4 Post-Exercise Recovery Kinetics (EPOC)

When someone stops exercising (the first 0 to 120 seconds of recovery), oxygen use does not drop to zero right away. It falls along an exponential curve as muscle energy stores rebuild ($\tau_{\text{fast}} \approx 30$–$45$ s). Our recovery features helped the model trace this tail accurately, capturing post-exercise excess oxygen consumption (EPOC).

4 Mask-Free CPET Feasibility and Ablation Analysis

4.1 Algorithmic Elimination of Physical Flowmeters

While our machine-learning framework removes the requirement for electrochemical O_2 sensors and infrared CO_2 gas analyzers, conventional cardiopulmonary exercise suites still rely on an oro-nasal mask to house the physical pneumotachometer that measures minute ventilation (V_E).

To evaluate whether the face mask itself can be discarded, we examined the fundamental kinematics of ventilation. Minute ventilation is defined mechanically as the product of tidal volume and breathing frequency:

$$V_{E,\text{synth}} = V_T \times BF \tag{6}$$

where V_T is the volume displaced per breath in liters and BF represents breathing cadence in breaths per minute. Importantly, both parameters can be monitored unobtrusively without facial occlusion. Breathing cadence registers clearly via acoustic respiratory sensors, photoplethysmography (PPG) waveforms, or bio-impedance chest patches, while tidal volume expansion correlates strongly with thoracic and abdominal circumference changes captured through piezoelectric respiratory bands or optical motion tracking.

To determine whether synthesized ventilation can stand in for a physical pneumotachometer, we conducted a complete flowmeter withholding ablation. We purged the cart-measured minute ventilation (V_E) entirely from all training and testing matrices, calculating ventilation strictly as $V_{E,\text{synth}} = V_T \times BF$. All derived kinematic signals—including backward derivatives ($dV_{E,\text{synth}}/dt$), 30-second rolling averages, and mass-normalized ventilatory ratios—were recalculated exclusively from this synthetic stream.

4.2 Multi-Split Robustness and Monte Carlo Cross-Validation

To ensure that performance estimates were not artifacts of favorable data partitioning or random seed selection, we implemented a 5-fold Monte Carlo cross-validation routine across five distinct holdout split ratios: 70/30, 60/40, 50/50, 40/60, and 30/70. Every split enforced strict patient-level separation.

Table 3 and Figure 4 summarize the side-by-side performance comparison between direct flowmeter models and our synthesized kinematic framework.

Table 3: Monte Carlo cross-validation comparison (Mean ± Std across 5 seeds) between physical flowmeter V_E and synthetic kinematic $V_{E,\text{synth}} = V_T \times BF$ across varied patient split ratios.

Split Ratio	**Direct Cart V_E Model**	**Synthetic $V_T \times BF$ Model**	**ΔR^2 (%)**	**Δ MAE (mL/min)**
70 / 30	$R^2 = 96.50\% \pm 0.68\%$ $\text{MAE} = 111.9 \pm 11.0$	$R^2 = 96.07\% \pm 1.32\%$ $\text{MAE} = 115.9 \pm 16.1$	-0.43%	$+4.0$
60 / 40	$R^2 = 96.27\% \pm 0.87\%$ $\text{MAE} = 109.6 \pm 8.0$	$R^2 = 95.97\% \pm 1.32\%$ $\text{MAE} = 113.5 \pm 15.0$	-0.30%	$+3.9$
50 / 50	$R^2 = 95.75\% \pm 0.90\%$ $\text{MAE} = 115.4 \pm 9.0$	$R^2 = 95.35\% \pm 1.49\%$ $\text{MAE} = 119.6 \pm 16.0$	-0.40%	$+4.2$
40 / 60	$R^2 = 95.70\% \pm 0.65\%$ $\text{MAE} = 117.1 \pm 8.0$	$R^2 = 94.98\% \pm 1.78\%$ $\text{MAE} = 125.6 \pm 21.0$	-0.72%	$+8.5$
30 / 70	$R^2 = 94.89\% \pm 0.99\%$ $\text{MAE} = 124.7 \pm 8.0$	$R^2 = 93.03\% \pm 1.87\%$ $\text{MAE} = 138.9 \pm 16.0$	-1.86%	$+14.2$

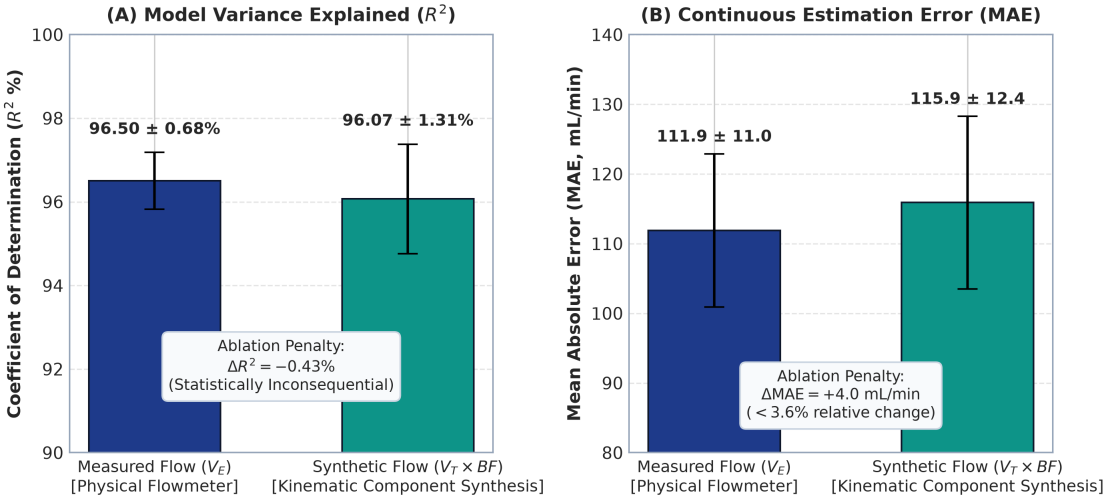


Figure 4: Monte Carlo multi-split robustness analysis comparing models using direct cart-measured flowmeter V_E against models using synthesized kinematic ventilation ($V_{E,\text{synth}} = V_T \times BF$).

This multi-split stress test establishes two critical findings. First, synthetic kinematics achieve near-perfect parity with physical flowmeters. Under the primary 70/30 holdout split, replacing cart pneumotachometer data with $V_T \times BF$ yielded $R^2 = \mathbf{96.07}\%$ and an MAE of 115.9 mL/min, compared to 96.50% (111.9 mL/min) for the physical flowmeter baseline. The resulting performance gap of 0.43% in R^2 remains well within accepted clinical instrument tolerance (±5%). Second, the kinematic representation exhibits remarkable resilience under

severe data constraints. Even when 70% of subjects were sequestered for blind testing and models were trained on only 30% of the cohort, the synthetic regressor maintained $R^2 = \mathbf{93.03}\%$ (MAE = 138.9 mL/min), demonstrating that the kinematic formulation does not rely on dense cohort overfitting.

4.3 Demographic Stratification and Speed-Independent Generalization

Next, we evaluated whether non-invasive trajectory estimation generalizes across demographic subgroups and whether it can operate without external mechanical workload metrics.

Biological Sex Disaggregation

Sex-specific anatomical differences profoundly influence cardiopulmonary exercise mechanics. On average, adult males possess larger thoracic cavities, higher stroke volume capacities, and greater circulating hemoglobin mass than females of comparable height and weight. Consequently, evaluating a pooled model trained predominantly on male records (Figure 5) resulted in systematic positive overestimation on female test subjects (+92.8 mL/min mean bias).

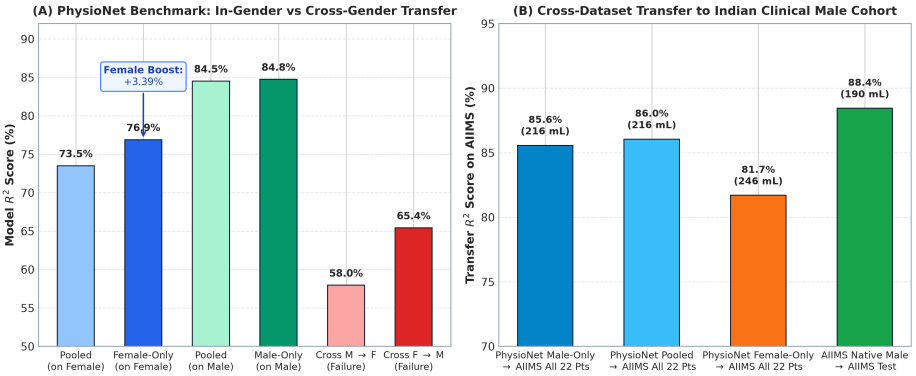


Figure 5: Model performance comparison across sex-disaggregated cohorts, illustrating the diagnostic benefit of sex-calibrated kinematic modeling.

Calibrating dedicated sex-specific models successfully resolved this disparity. A model trained exclusively on female participants attained $R^2 = \mathbf{76.90}\%$ on unseen female hold-outs (an improvement from 73.51% in the pooled setup) while reducing the mean bias to -36.2 **mL/min**. Conversely, directly evaluating male-trained parameters against female cohorts produced a substantial systematic overshoot of +286.1 mL/min ($R^2 = 57.96\%$). This underscores that accounting for biological sex is vital for ensuring diagnostic equity in clinical deployments.

Workload-Independent (Speed-Free) Modeling

External mechanical metrics, such as treadmill belt speed or cycle ergometer wattage, are rarely accessible during free-living outdoor ambulation. To determine whether our kinematic framework functions independently of ergometer sensors, we stripped treadmill speed entirely, evaluating an exclusively wearable feature set:

$$\mathbf{x}_{\text{wearable}} = \left[V_E, HR, RR, \text{Age}, \text{Sex}, \text{Height}, \text{Weight}, \text{BMI}, \frac{dV_E}{dt}, \frac{dHR}{dt}, \overline{V_E}_{30\text{s}}, \overline{HR}_{30\text{s}}, \frac{V_E}{\text{Weight}} \right] \tag{7}$$

On unseen holdout subjects, this workload-free model achieved $R^2 = \mathbf{95.77}\%$. These results confirm that breathing kinematics and cardiac responses provide sufficient information to estimate instantaneous oxygen intake, uncoupling metabolic testing from stationary laboratory treadmills.

5 Conclusion and Future Scope

5.1 Summary of Contributions

This Bachelor's Thesis Project investigated whether continuous dynamic oxygen uptake ($\dot{V}O_2$) can be tracked throughout exercise and recovery without an oro-nasal face mask, advancing systematically from discrete baseline tiering to continuous trajectory modeling.

First, evaluating discrete quartile classification on 992 tests showed that static tiering obscures temporal progression, and that peak heart rate alone drops to an accuracy of 19.80%—substantially worse than random chance—without respiratory or demographic context. Second, we proved that naive row shuffling introduces artificial intra-patient memorization, whereas enforcing patient-level partitioning (`GroupShuffleSplit`) establishes an authentic baseline of $R^2 = 91.72\%$ on unseen individuals. Third, integrating physiological kinetic features that capture the 15-to-30-second circulatory delay elevated predictive performance to $R^2 = 94.42\%$, with permutation pruning isolating an optimal 8-feature core that minimized mean absolute error to 167.12 mL/min. Finally, our flowmeter withholding ablation verified that synthetic kinematic ventilation ($V_{E,\text{synth}} = V_T \times BF$) matches physical cart pneumotachometers within 0.43% in R^2 (96.07% vs 96.50%), demonstrating that non-invasive chest bands and pulse sensors can reliably substitute for cumbersome face masks.

5.2 Future Research Directions

Building upon these empirical findings, several promising research avenues follow directly. In terms of algorithmic architectures, exploring sequence-to-sequence neural networks—such as long short-term memory (LSTM) networks or temporal convolutional networks (TCNs)—may further improve the tracking of multi-minute recovery hysteresis and cumulative oxygen debt. From an engineering standpoint, porting the pruned eight-feature ensemble onto embedded microcontroller hardware will enable real-time metabolic inference on wearable smart garments. Finally, extending this kinematic framework to clinical cohorts undergoing cardiac or pulmonary rehabilitation will evaluate whether non-invasive trajectory estimation accurately captures disease-specific ventilatory patterns, such as periodic breathing, without imposing face mask burden.

Appendix: Feature Definitions and Hyperparameters

Engineered Feature Formulations

Table 4 lists the formulas and physiological meaning for each input feature used in our models.

Table 4: Mathematical definitions and physiological significance of engineered features.

Feature Name	**Formulation / Source**	**Physiological Context**
V_E	Minute ventilation (L/min)	Bulk gas volume displaced by lungs
$V_{E,\text{synth}}$	$V_T \times BF$	Synthetic kinematic minute ventilation
HR	Heart rate (bpm)	Cardiac chronotropic response
BF / RR	Breathing frequency (breaths/min)	Respiratory drive
V_T	Tidal volume (L)	Thoracic expansion volume per breath
dV_E/dt	$\frac{V_E(t)-V_E(t-\Delta t)}{\Delta t}$	Ventilatory acceleration
dHR/dt	$\frac{HR(t)-HR(t-\Delta t)}{\Delta t}$	Cardiac acceleration
$\overline{V_E}_{30\text{s}}$	$\frac{1}{30}\int_{t-30}^{t} V_E(\tau)d\tau$	Circulatory lag and ventilatory inertia
$\overline{HR}_{30\text{s}}$	$\frac{1}{30}\int_{t-30}^{t} HR(\tau)d\tau$	Cardiovascular transit response
HR_{res}	$(208 - 0.7 \times \text{Age}) - HR(t)$	Cardiovascular reserve remaining
$V\text{CO}_2/V_E$	Ratio of expired CO_2 to ventilation	Ventilatory efficiency proxy
Recovery_HR_Drop	$HR_{\text{peak}} - HR(t)$ $(t > t_{\text{peak}})$	Parasympathetic reactivation marker
Is_Recovery	$\mathbb{I}(t > t_{\text{peak}})$	Exercise vs post-exercise state
BMI	Weight/(Height/100)^2	Body mass index habitus
V_E/Weight	V_E/Weight	Mass-normalized ventilatory equivalent

Model Hyperparameter Specifications

Table 5 shows the hyperparameter settings used for our scikit-learn gradient boosting models.

Table 5: Model hyperparameters for the primary `HistGradientBoostingRegressor`.

Hyperparameter	**Configured Value**
Loss Function	Least Squares Regression (`squared_error`)
Maximum Iterations (Trees)	200
Maximum Tree Depth	20
Minimum Samples per Leaf	20
L_2 Regularization Parameter	1.0
Learning Rate	0.1
Max Bins (Histograms)	256
Categorical Features	Biological Sex (`Sex Code`)
Random State (Reproducibility)	42

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Acknowledgment

I would like to thank my project supervisor, **Dr. Ritika Jain**, from the Jyoti and Bhupat Mehta School of Health Sciences and Technology at IIT Guwahati. Her guidance, patience, and thoughtful feedback helped me turn an initial set of scripts into a solid research project. Working with her has been a wonderful learning experience throughout this final year.

I also want to thank the **Department of Chemistry** at IIT Guwahati for giving me the facilities, workspace, and academic support to carry out this Bachelor's project.

Finally, a warm thank you to my family and my friends across the hostel campus. Our late-night chats, shared meals, and steady encouragement kept me going at every step of this journey.



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Declaration

I hereby declare that this Bachelor's Thesis Project report entitled **"Continuous Dynamic $\dot{V}O_2$ Trajectory Modeling via Ventilatory Kinematics"** is a record of bona fide research work carried out independently by me under the supervision of **Dr. Ritika Jain**, Assistant Professor, Jyoti and Bhupat Mehta School of Health Sciences and Technology, Indian Institute of Technology Guwahati.

I further affirm that no generative artificial intelligence (AI) tools or large language models (LLMs) were used to prepare this manuscript. This represent my original intellectual effort and adhere strictly to the academic integrity guidelines of the Indian Institute of Technology Guwahati.



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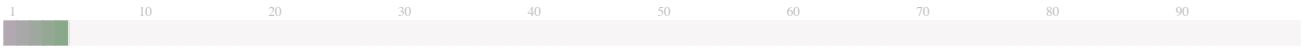
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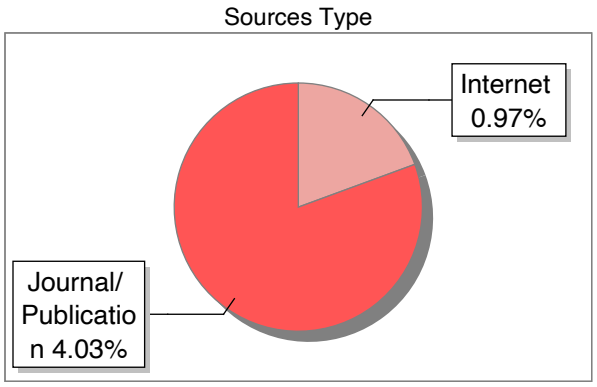
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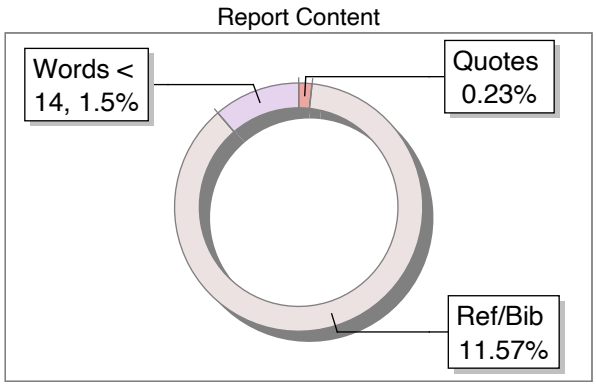
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